Hydraulic-Press — Maintenance SOP (Faultiest Station on Line-B)

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> **Audience:** Maintenance Tech (primary) · Line Manager **Applies to:** Line-B, station 2, Hydraulic-Press

1. Purpose

The Hydraulic-Press deep-draws the blank from station 1 into the 3-D housing shape. With a 1.5 % per-cycle fault probability — the **highest on Line-B** — this press is the single most common source of downtime on the sheet-metal line. This SOP exists to give maintenance technicians a fast, predictable path from a `Fault` telemetry event back to `Running`.

2. Normal operating envelope

- Ideal cycle time: **12 s**
- Max acceptable cycle time: **18 s**
- Expected reject rate: **≤ 4 %**
- Fault probability per cycle: **1.5 %** (line max)
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Blanking-Press (station 1) is not feeding blanks.	Verify Blanking-Press state.
`Blocked`	Trimming-Station (station 3) cannot accept the formed part.	Verify Trimming state.

Because of this press's high fault rate, **most downstream `Starved` events on Line-B trace back to this station**. When triaging starvation on stations 3 or 4, check this press first.

4. Faults & corrective actions

4.1 `Pressure-Loss`

- **Symptoms in telemetry:** `fault_type = "Pressure-Loss"`. The press cannot complete the down stroke; cycle time often spikes to the 18 s max on the last successful cycle.
- **Likely root cause:** internal pump leak; relief valve drift; hose burst.
- **Corrective action:**
  1. LOTO and bleed stored pressure per `Safety_Lockout_Tagout_Procedure.pdf`.
  2. Check fluid level in the reservoir; top off only with the OEM-spec hydraulic fluid.
  3. Inspect hoses and fittings for visible leaks; replace as needed.
  4. Pressure-test to system rating (typ. 3,000 psi); verify hold for 5 minutes with the press at top of stroke.
- **Restart criteria:** rated pressure held for 5 minutes, no leaks.

4.2 `Seal-Failure`

- **Symptoms in telemetry:** `fault_type = "Seal-Failure"`. Often follows a period of slow pressure decay and visible weeping around the main ram.
- **Likely root cause:** main ram seal has hardened or cut. End of service life is typically 18–24 months under normal duty.
- **Corrective action:**
  1. LOTO. Drop the ram with overhead crane onto blocks.
  2. Replace the main ram seal kit (rod seal + wiper + back-up ring).
  3. Re-flush the line with fresh hydraulic fluid.
  4. Cycle the press 20 times at half-rated pressure before resuming production.
- **Restart criteria:** no weep across a 30-minute observation window.

4.3 `Valve-Fault`

- **Symptoms in telemetry:** `fault_type = "Valve-Fault"`. Erratic stroke timing, or the press fails to return.
- **Likely root cause:** directional control valve solenoid burned out, or spool stuck from contamination.
- **Corrective action:**
  1. LOTO.
  2. Inspect the directional valve solenoid for continuity; replace coil.
  3. If the spool is stuck, replace the valve cartridge.
  4. Sample-test the hydraulic fluid for ISO 18/16/13 cleanliness; flush if out of spec.
- **Restart criteria:** ten consecutive cycles within ideal cycle time window.

5. Preventive maintenance schedule

- **Daily:** fluid level check; visual leak walk-around.
- **Weekly:** filter element delta-pressure check.
- **Monthly:** sample hydraulic fluid for particulate count.
- **Quarterly:** main ram seal inspection (record service hours).
- **Annually:** full pump and motor service.

A missed quarterly seal inspection escalates to the Plant Manager per `OEE_Targets_and_Escalation.pdf` because of this press’s outsized impact on line OEE.

6. References

- Maintenance_Workflow_Overview.pdf
- Safety_Lockout_Tagout_Procedure.pdf
- OEE_Targets_and_Escalation.pdf
- Line-B_01_Blanking-Press_SOP.pdf
- Line-B_03_Trimming-Station_SOP.pdf